

Primetrade.ai AI Internship Task

Trader Performance vs Bitcoin Market Sentiment

Candidate: K N Chetan | Dataset: Hyperliquid Historical Trader Data + Bitcoin Fear & Greed Index

Objective. Explore how trader performance changes across Bitcoin market sentiment regimes and convert the findings into practical trading strategy insights.

Metric	Value
Rows analyzed	211,224
Realized PnL rows	104,408
Accounts / Coins	32 / 246
Trading date range	2023-05-01 to 2025-05-01
Total gross PnL	\$10,296,959
Total fees	\$245,858
Total net PnL	\$10,051,101
Overall realized win-rate	82.30%
Overall profit factor	4.42
Overall PnL per \$1k notional	8.44

1. Data Quality and Assumptions

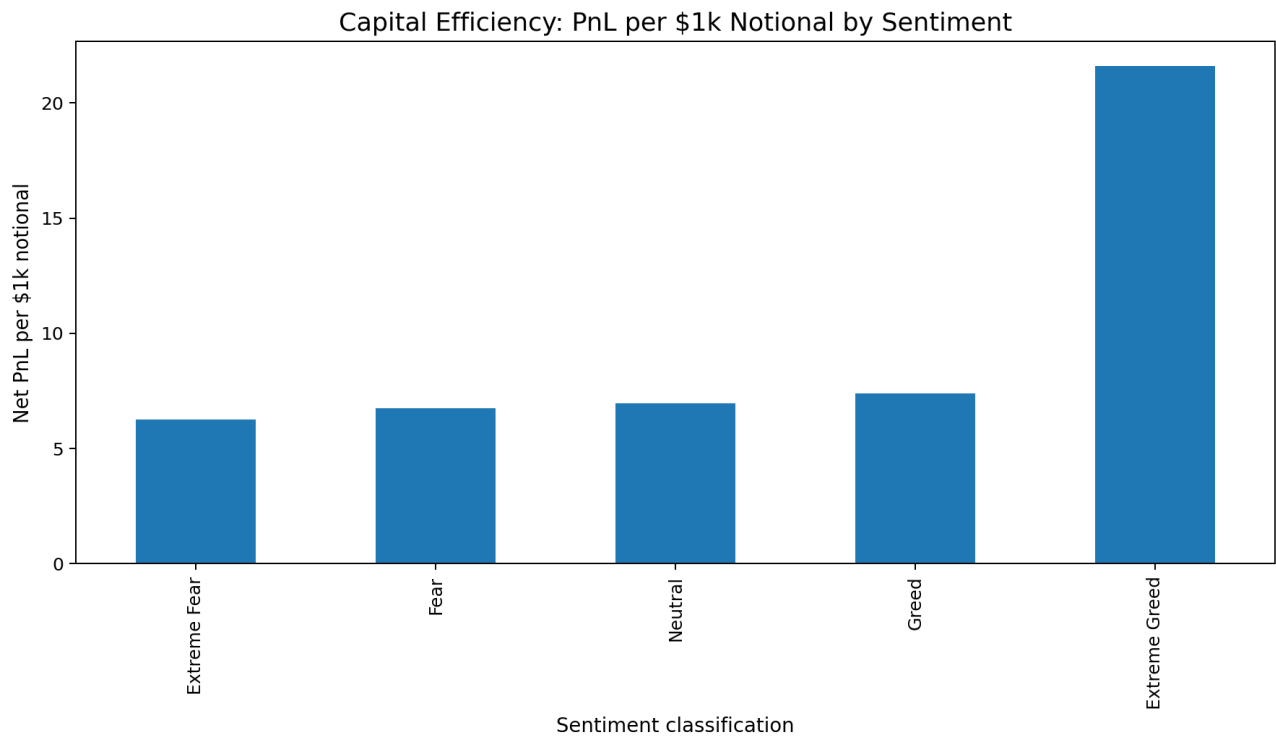
The historical file contains fill/activity rows. Because many rows have Closed PnL equal to zero, the analysis separates trade activity from realized outcome rows. Win-rate, average realized PnL, and profit factor are calculated only where Closed PnL is non-zero. Net PnL is calculated as Closed PnL minus Fee.

The Fear & Greed file was reindexed to a continuous calendar across the trading period. Missing sentiment was forward-filled; this affected 1 calendar date, including 2024-10-26. This has negligible row impact but makes the merge auditable.

2. Sentiment Scorecard

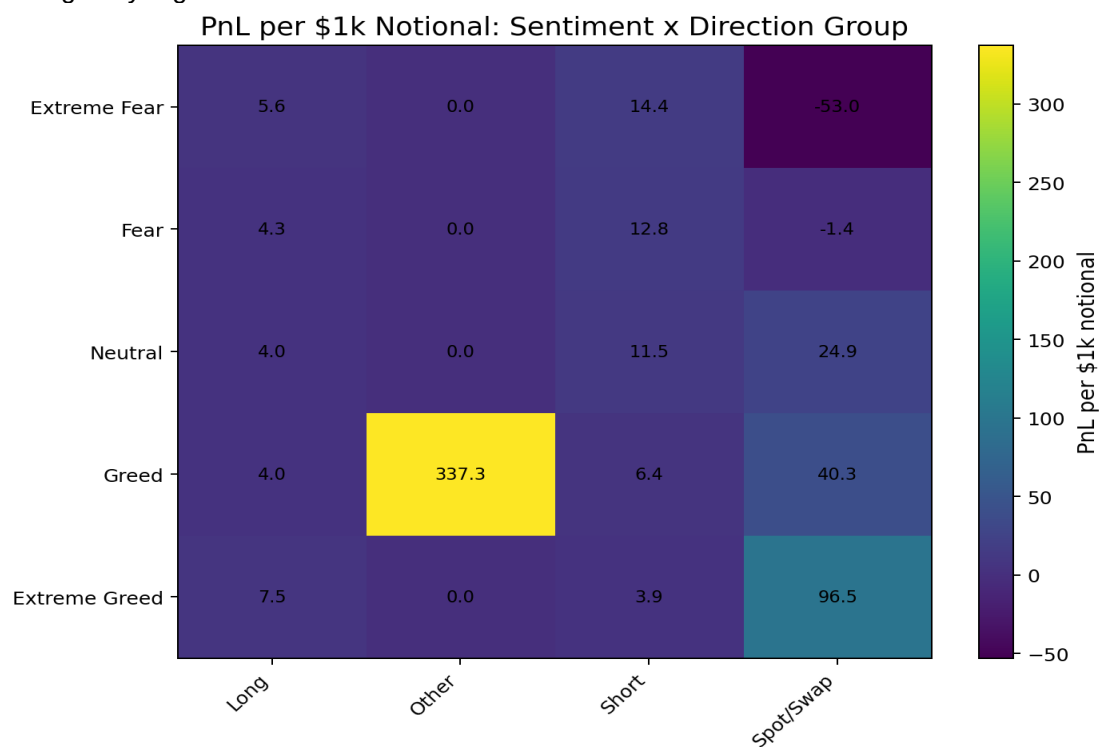
Extreme Greed produced the strongest capital efficiency, while Fear generated the largest absolute net PnL due to higher notional and realized trade count. Greed was profitable but less efficient than Extreme Greed.

Sentiment	Rows	Realized	Notional	Net PnL	Win Rate	PF	PnL/\$1k
Extreme Fear	21,400.00	10,406.00	114,484,261.44	715,221.61	75.77%	2.14	6.25
Fear	61,837.00	29,808.00	483,324,789.79	3,264,698.49	85.37%	6.50	6.75
Neutral	37,686.00	18,159.00	180,242,063.08	1,253,546.41	82.02%	4.24	6.95
Greed	50,309.00	25,182.00	288,671,163.58	2,129,494.06	76.17%	3.03	7.38
Extreme Greed	39,992.00	20,853.00	124,465,164.57	2,688,140.65	88.79%	10.90	21.60



3. Direction and Behavior Patterns

Regime-dependent behavior is visible. Spot/Swap-like Sell activity was strongest in Extreme Greed, while Spot/Swap activity was negative in Fear and Extreme Fear. Long/Short groups remained profitable overall, but efficiency changed by regime.



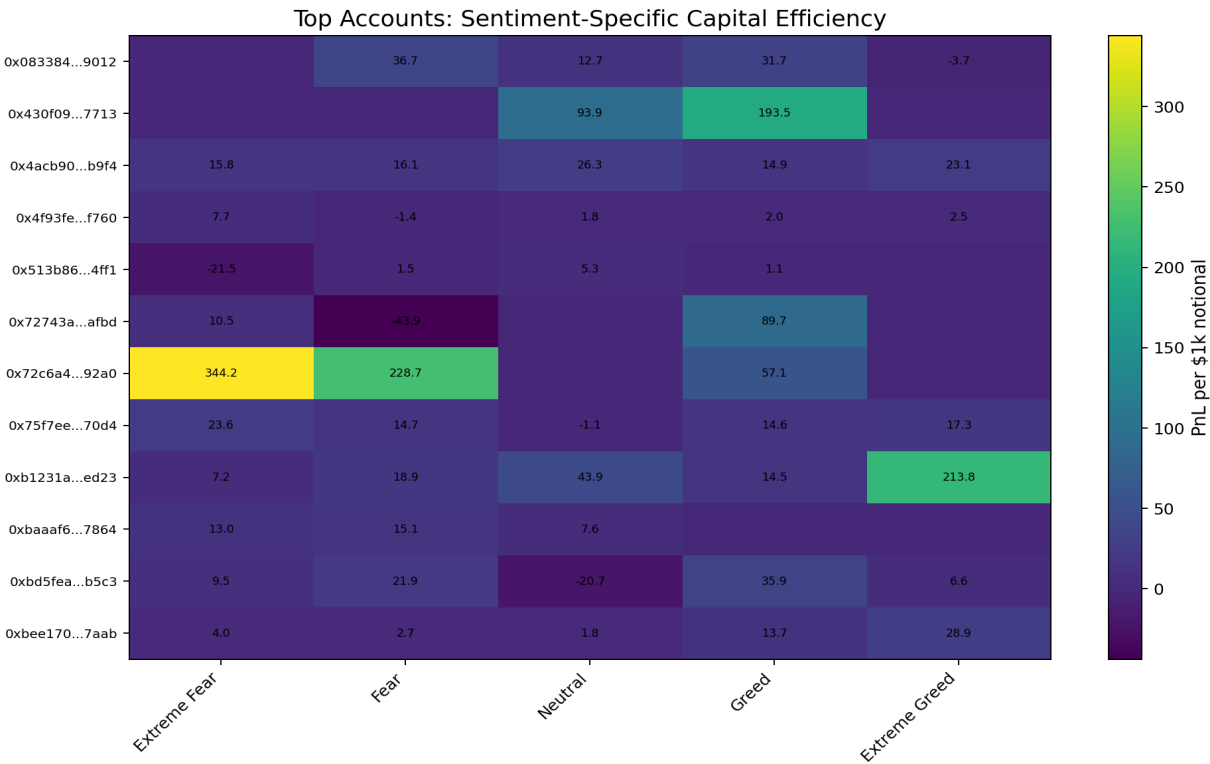
Sentiment	Direction	Realized	Net PnL	Win Rate	PF	PnL/\$1k
Greed	Other	9.00	57,449.24	88.89%	1,967.40	337.35
Extreme Greed	Spot/Swap	7,161.00	2,079,233.64	92.47%	72.14	96.47
Greed	Spot/Swap	5,846.00	763,832.53	78.94%	6.50	40.29
Neutral	Spot/Swap	2,298.00	208,301.52	77.28%	7.58	24.94
Extreme Fear	Short	3,117.00	379,113.04	69.65%	2.41	14.40
Fear	Short	9,212.00	1,883,628.33	84.59%	13.21	12.76

Sentiment	Direction	Realized	Net PnL	Win Rate	PF	PnL/\$1k
Neutral	Short	5,849.00	543,105.51	77.16%	4.01	11.45
Extreme Greed	Long	7,203.00	430,737.22	88.23%	17.57	7.46
Greed	Short	11,326.00	632,810.19	67.94%	1.75	6.41
Extreme Fear	Long	6,252.00	477,383.00	84.45%	3.32	5.58

4. Trader Specialization

Trader-level results are not uniform. Some accounts perform best in Fear or Extreme Greed, while others lose money in the same regimes. This supports a sentiment-aware capital allocation layer rather than one global trading rule.

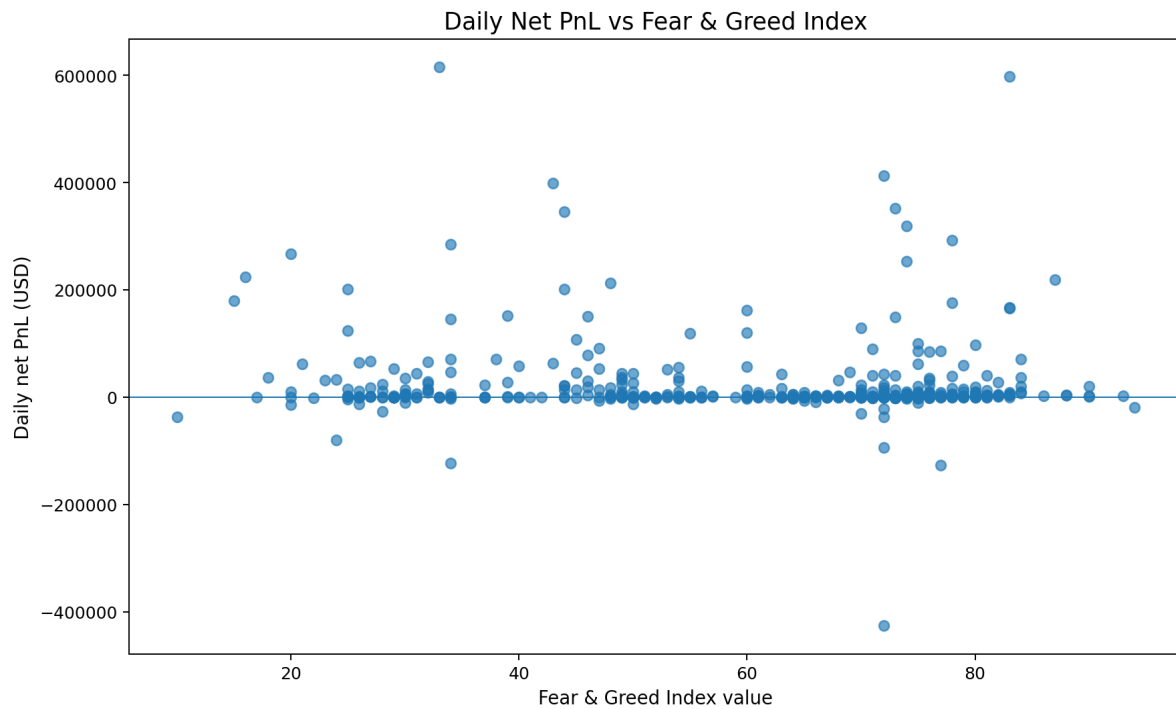
Account	Sentiment	Realized	Net PnL	Win Rate	PF	PnL/\$1k
0x083384...9012	Fear	1,051.00	1,110,127.70	89.06%	14.88	36.68
0xb1231a...ed23	Extreme Greed	893.00	1,103,779.16	93.73%	139.35	213.85
0xbaaaf6...7864	Fear	6,283.00	615,230.50	95.53%	10,599.03	15.10
0xb1231a...ed23	Greed	2,464.00	525,962.20	59.90%	18.94	14.50
0xbee170...7aab	Extreme Greed	4,520.00	477,137.31	88.54%	18.37	28.87
0x72743a...afbd	Greed	269.00	453,028.05	69.89%	74.56	89.71
0xb1231a...ed23	Neutral	1,535.00	398,266.55	85.08%	31.26	43.94
0x513b86...4ff1	Neutral	1,443.00	367,919.84	96.67%	12.73	5.28
0x513b86...4ff1	Fear	2,576.00	329,571.05	83.54%	7.79	1.46
0x4acb90...b9f4	Fear	599.00	293,174.58	95.33%	312.51	16.08



5. Statistical Validation

The realized PnL distributions and realized win-rates differ by sentiment regime. However, the daily correlation between Fear & Greed value and total daily net PnL is weak and not statistically significant. This means sentiment should be treated as a regime/risk feature, not as a standalone prediction engine.

Test	Statistic	P-value	Effect	Interpretation
Kruskal-Wallis	749.52	0.00	0.01	PnL differs by regime; effect small due heavy tails.
Chi-square	1,749.18	0.00	0.13	Win-rate differs across regimes.
Spearman correlation	0.04	0.34	0.04	Daily FGI-PnL link is weak/non-significant.



6. Predictive Sanity Check

A simple out-of-time logistic regression check was used to test whether sentiment alone predicts profitable realized rows. The result was weak, which supports a cautious interpretation: sentiment is useful as a conditional overlay with account, direction, size, and instrument behavior, not as an isolated signal.

Model	Test Rows	Test Win Rate	ROC AUC	Avg Precision	Accuracy
sentiment_only	20882	72.34%	0.48	0.74	0.53
trade_context_no_sentiment	20882	72.34%	0.51	0.71	0.54
trade_context_plus_sentiment	20882	72.34%	0.49	0.72	0.54

7. Actionable Trading Strategy Insights

Regime filter: Use Fear & Greed as a risk overlay that changes size, trader allocation, and preferred direction group.

Extreme Greed: Best capital efficiency in this dataset. Spot/Swap-like sell activity and selected high-performing accounts deserve higher allocation, subject to live risk controls.

Fear: Large absolute PnL, especially in short-close and long-close groups, but efficiency is lower than Extreme Greed. Favor accounts with proven Fear-regime performance.

Greed: Profitable but mixed. Apply stricter account-level filters and cap exposure for historically weak Greed-regime accounts.

Risk control: Use PnL per \$1k notional and profit factor together. Absolute PnL alone can be misleading because Fear had much higher notional.

Next step: Convert this into a trader allocation matrix: sentiment regime x account x direction x size bucket, with minimum realized-trade thresholds.

Appendix: Output Files

The project folder includes the notebook, this report, charts, scorecard tables, statistical test results, and a compact merged data sample. The raw CSVs are included so the notebook is reproducible without extra setup.